

4. HOSPITAL PATIENT ANALYSIS DASHBOARD

Build a dashboard using patient data containing age, blood pressure, cholesterol, BMI, treatment cost, and admission date. Present distributions, counts, and relationships between health-related variables using Matplotlib. Include time-based plots to analyze admissions and use lag/autocorrelation plots to identify recurring admission patterns.

SOLUTION:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

np.random.seed(42)

n = 120

dates = pd.date_range("2026-01-01", periods=180, freq="D")

admission_dates = pd.to_datetime(

    np.random.choice(dates, n, replace=True)

)

patients = pd.DataFrame({

    "Age": np.random.randint(18, 81, n),

    "BloodPressure": np.clip(

        np.random.normal(128, 18, n), 90, 190

    ),

    "Cholesterol": np.clip(

        np.random.normal(205, 35, n), 120, 320

    ),

    "BMI": np.clip(

        np.random.normal(26, 4.5, n), 16, 42

    ),

    "AdmissionDate": admission_dates

})

patients["TreatmentCost"] = np.clip(

    12000

    + patients["Age"] * 450

    + patients["BMI"] * 800

    + patients["BloodPressure"] * 100

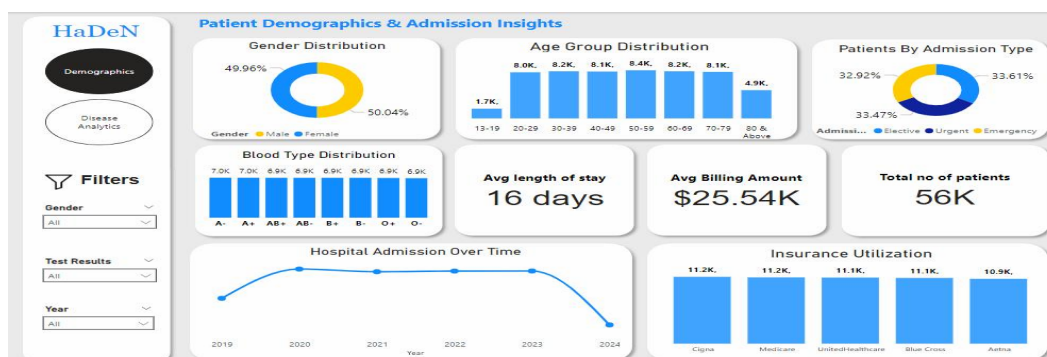
    + np.random.normal(0, 7000, n),
```

```

5000,
90000
)
patients = patients.sort_values("AdmissionDate")
daily = (
    patients
    .set_index("AdmissionDate")
    .resample("D")
    .size()
)
daily = daily.asfreq("D", fill_value=0)
lags = list(range(1, 15))
autocorrelation = [
    daily.autocorr(lag=i)
    for i in lags
]
fig, ax = plt.subplots(3, 3, figsize=(16, 11))
fig.suptitle(
    "Hospital Patient Analysis Dashboard",
    fontsize=20,
    fontweight="bold"
)
ax[0, 0].hist(
    patients["Age"],
    bins=10,
    edgecolor="black"
)
ax[0, 0].set_title("Age Distribution")
ax[0, 0].set_xlabel("Age")
ax[0, 0].set_ylabel("Patients")
ax[0, 1].hist(
    patients["BloodPressure"],

```

OUTPUT:



5. STOCK MARKET PATTERN ANALYSIS DASHBOARD

Develop a dashboard using historical stock data containing date, opening price, closing price, high, low, and trading volume. Visualize price evolution over time, distributions, and relationships between price and volume. Use Pandas plotting, scatter plots, lag plots, autocorrelation plots, and bootstrap plots to investigate trends, dependencies, and uncertainty in stock behavior.

SOLUTION:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

np.random.seed(10)

n = 180

dates = pd.bdate_range(

    "2026-01-02",

    periods=n

)

returns = np.random.normal(

    0.001,

    0.018,

    n

)

close = 100 * np.exp(

    np.cumsum(returns)

)

open_price = close * (

    1 + np.random.normal(0, 0.006, n)

)

high = np.maximum(

    open_price,

    close

) * (

    1 + np.random.uniform(

        0.002,

    )

)

ax[2, 0].set_title(

    "Return Autocorrelation"

)

ax[2, 0].set_xlabel(

    "Lag"

)

ax[2, 0].set_ylabel(

    "Autocorrelation"

)

ax[2, 1].hist(

    bootstrap_means,
```

```

bins=30,

edgecolor="black"

)

ax[2, 1].axvline(

    lower,

    linestyle="--",

    linewidth=2

)

ax[2, 1].axvline(

    upper,

    linestyle="--",

    linewidth=2

)

ax[2, 1].axvline(

    returns_data.mean(),

    linewidth=2

)

ax[2, 1].set_title(

    "Bootstrap Mean Return"

)

ax[2, 1].set_xlabel(

    "Mean Daily Return"

)

ax[2, 1].set_ylabel(

    "Frequency"

)

stock["Volume"].plot(

    ax=ax[2, 2]

)

ax[2, 2].set_title(

    "Trading Volume Over Time"

```

OUTPUT:

